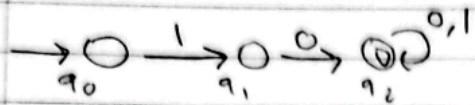


NFA design exercise 1

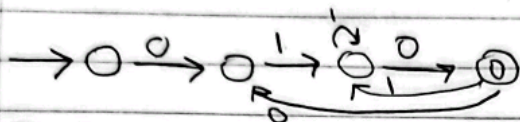
Yostina SeAddik

6) {string s | s starts with 10}

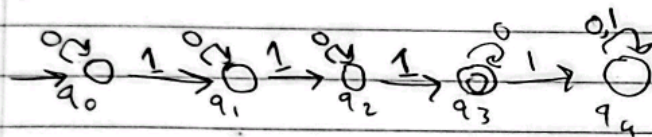


current state	0	1
q_0	$\{ \}$	$\{q_1\}$
q_1	$\{q_2\}$	$\{ \}$
q_2	$\{q_2\}$	$\{q_2\}$

8) {string s | s starts with 01 and end with 10}

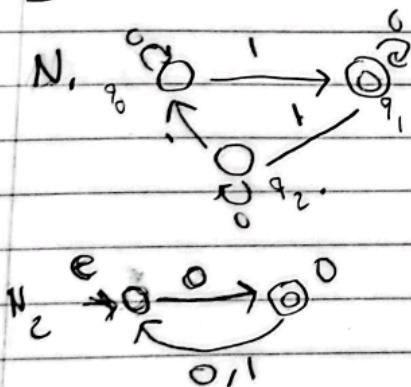


12) {string s | s contains exactly 3 1's}

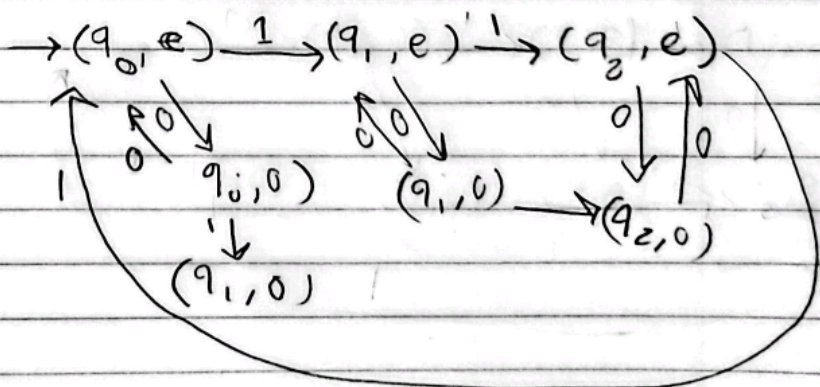


current	0	1
$\rightarrow q_0$	$\{q_0\}$	$\{q_1\}$
q_1	$\{q_1\}$	$\{q_2\}$
q_2	$\{q_2\}$	$\{q_3\}$ = final
q_3	$\{q_3\}$	$\{q_4\}$
q_4	$\{q_4\}$	$\{q_4\}$

21) {string s | s has $3k+1$ of 1's, where $k \geq 0$ & odd numbers of 0's}



$N_1 \cap N_2$



24) {string s / s has odd # of 0's or $s=001$ }

